

SO(3) Equivariant Neural Networks

Lecture 2

Features Have Geometric Types

Representations, irreps, spherical components, and Wigner matrices

90-minute journal-club lecture note

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Companion notebook included in the series package

Session map and learning goals

hook	representations	multipoles	break	AI features	notebook	discussion
0			45			90 min

By the end of this lecture, you should be able to:

- explain a representation as a rule that assigns a matrix to each rotation;
- interpret $\ell = 0, 1, 2$ as scalar, vector, and quadrupolar feature types;
- distinguish an irrep's magnetic index m from a neural-network channel index;
- convert a vector between Cartesian and spherical components;
- explain what the Wigner matrix $D^{(\ell)}(R)$ does.

Key idea

A neural network feature should not be viewed as an anonymous list of numbers. Its type specifies how its components must mix when the physical system is rotated.

1 From a physical rotation to a representation

The same physical rotation R acts differently on different objects:

$$s' = s, \quad v' = Rv, \quad T' = RTR^T. \quad (1)$$

A *representation* is a consistent matrix realization of this action. It assigns a matrix $D(R)$ to each $R \in \text{SO}(3)$ and satisfies

$$D(R_1)D(R_2) = D(R_1R_2), \quad D(I) = I. \quad (2)$$

The representation law ensures that the description respects composition of physical rotations.

Math lens

A representation acts on a vector space. Different representations may have different dimensions, but they describe the same group. The ordinary 3×3 rotation matrix is only one representation of $\text{SO}(3)$.

2 Irreducible representations of $\text{SO}(3)$

A representation may contain smaller subspaces that never mix under rotation. Decomposing until no further rotation-preserving split is possible yields *irreducible representations*, or irreps. For $\text{SO}(3)$, irreps are labeled by

$$\ell = 0, 1, 2, \dots, \quad \dim V_\ell = 2\ell + 1. \quad (3)$$

$\ell = 0$		$2\ell + 1 = 1$ components
$\ell = 1$		$2\ell + 1 = 3$ components
$\ell = 2$		$2\ell + 1 = 5$ components
$\ell = 3$		$2\ell + 1 = 7$ components

The first three types carry immediate physical interpretations:

ℓ	Dimension	Geometric type	Typical physics example
0	1	scalar	charge, energy, radial feature
1	3	vector	position, force, dipole moment
2	5	symmetric-traceless tensor	quadrupole, anisotropy

2.1 Why a symmetric-traceless tensor has five components

A symmetric 3×3 matrix has six independent components. Its trace is a one-dimensional scalar piece. Removing the trace leaves a five-dimensional symmetric-traceless piece:

$$6 = 1 + 5. \quad (4)$$

Under rotations, these pieces never mix. They are the $\ell = 0$ and $\ell = 2$ parts.

Physics lens

The sequence $\ell = 0, 1, 2, \dots$ is also the multipole hierarchy: monopole, dipole, quadrupole, and higher angular patterns. Increasing ℓ lets a model describe finer directional structure.

3 The spherical basis (ℓ, m)

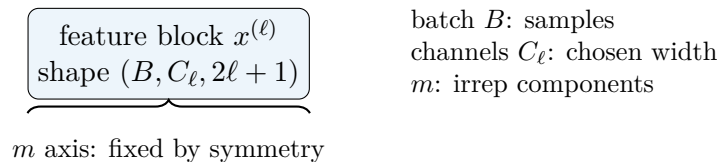
Inside an irrep ℓ , components are indexed by

$$m = -\ell, -\ell + 1, \dots, +\ell. \quad (5)$$

We write a feature as $x_m^{(\ell)}$. For a vector,

$$x^{(1)} = \begin{pmatrix} x_{-1} & x_0 & x_{+1} \end{pmatrix}^\top. \quad (6)$$

The index m labels components within one geometric object. It is not a learnable channel.



3.1 Cartesian and spherical vector components

The notebook uses

$$v_{-1} = \frac{v_x - iv_y}{\sqrt{2}}, \quad v_0 = v_z, \quad v_{+1} = -\frac{v_x + iv_y}{\sqrt{2}}. \quad (7)$$

Writing $v_{\text{sph}} = U_1 v_{\text{cart}}$ gives

$$U_1 = \begin{pmatrix} 1/\sqrt{2} & -i/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & -i/\sqrt{2} & 0 \end{pmatrix}, \quad U_1 U_1^\dagger = I. \quad (8)$$

Because U_1 is unitary, no information is lost. The inverse is $v_{\text{cart}} = U_1^\dagger v_{\text{sph}}$.

Pause and discuss

Why might complex coordinates be useful even when all physical vectors are real? What property of a change of basis guarantees that physical predictions do not depend on the chosen basis?

3.2 The spherical reality condition

For a real physical spherical tensor, the components are related by

$$\left(x_m^{(\ell)}\right)^* = (-1)^m x_{-m}^{(\ell)}. \quad (9)$$

Thus complex components are not independent extra physical degrees of freedom. They are a convenient coordinate system adapted to rotations about the z axis and to angular-momentum coupling.

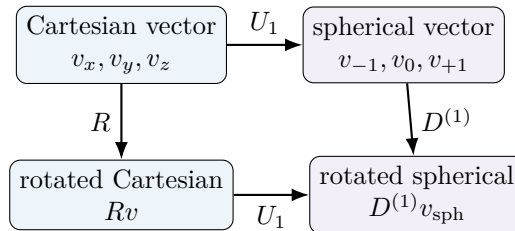
4 Wigner D matrices

The generalized rotation matrix for type ℓ is the Wigner matrix $D^{(\ell)}(R)$:

$$x_m'^{(\ell)} = \sum_{m'=-\ell}^{\ell} D_{mm'}^{(\ell)}(R) x_{m'}^{(\ell)}. \quad (10)$$

Its shape is $(2\ell + 1) \times (2\ell + 1)$. In particular,

$$D^{(0)}(R) = 1, \quad D^{(1)}(R) = U_1 R U_1^\dagger. \quad (11)$$



The diagram commutes because $D^{(1)} = U_1 R U_1^\dagger$. It is one physical rotation written in two bases, not two different rotations.

Equation (11) names only $\ell = 0$ and $\ell = 1$, which looks like a gap. It is not: those two already determine every other ℓ . The construction needs the coupling coefficients of the next lecture, so we return to it there — see Lecture 3, Section 5. For now it is enough to know that $D^{(2)}$, the only other Wigner matrix this course uses, is built for you in the notebook, and that nothing new has to be invented to reach it.

5 From representation theory to feature design

An equivariant network stores features by irrep type:

```
features = {
    0: scalars,      # shape (batch, C0, 1)
    1: vectors,      # shape (batch, C1, 3)
    2: quadrupoles,  # shape (batch, C2, 5)
}
```

Rotating the feature dictionary means applying the correct $D^{(\ell)}(R)$ to each final axis. The channel axis is unaffected: every channel of the same type follows the same transformation law.

AI lens

The irrep label is a type system for geometry. It prevents the network from accidentally adding a scalar to a vector, applying an unrelated matrix to vector components, or confusing channel width with representation dimension.

6 Notebook laboratory

Open `lecture_02_irreps_and_spherical_basis.ipynb`. You will:

1. implement Cartesian-to-spherical and inverse conversions;
2. perform a genuine round-trip check against the original vector;
3. construct $D^{(0)}$, $D^{(1)}$, and $D^{(2)}$;
4. verify unitarity and the representation law;
5. rotate the same vector through Cartesian and spherical routes;
6. watch an $\ell = 2$ object rotate, to see that five components turn as one rigid pattern.

7 Questions and discussion

Question 2.1. What is the difference between the physical rotation R and the Wigner matrix $D^{(\ell)}(R)$?

Question 2.2. For $\ell = 0, 1, 2, 3$, state the number of m components and give a physical interpretation when one is familiar.

Question 2.3. Explain the difference between the channel index and the magnetic index m . Which is chosen by the network designer?

Question 2.4. Why does the use of complex spherical components not double the physical degrees of freedom of a real vector?

Question 2.5. Show algebraically that $D^{(1)}(R) = U_1 R U_1^\dagger$ obeys the representation law if R does.

Question 2.6. A proposed round-trip test computes only the imaginary part of the reconstructed Cartesian vector. Why is that insufficient, and what should the test compute instead?

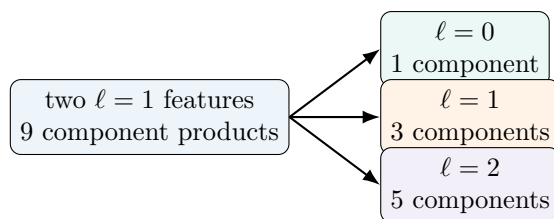
Checkpoint

You are ready for Lecture 3 if you can explain: ℓ names a geometric type, m labels components inside that type, and channels count how many learned features of that type are present.

Bridge to Lecture 3: when two feature types meet

Knowing how one feature rotates is only half of the architecture problem. A nonlinear model must also combine features. Two vectors contain nine pairwise component products, yet those products do not form one irreducible type. They separate into three rotational pieces:

$$(2 \cdot 1 + 1)(2 \cdot 1 + 1) = 9 = 1 + 3 + 5, \quad 1 \otimes 1 = 0 \oplus 1 \oplus 2. \quad (12)$$



Lecture 3 introduces the Clebsch–Gordan table that performs this reorganization while preserving the rotation law.